

Lesson 2: Linear combinations and spans

$$v_1, v_2, \dots, v_n \text{ in } \mathbb{R}^n$$

$$c_1 v_1 + c_2 v_2 + \dots + c_n v_n$$

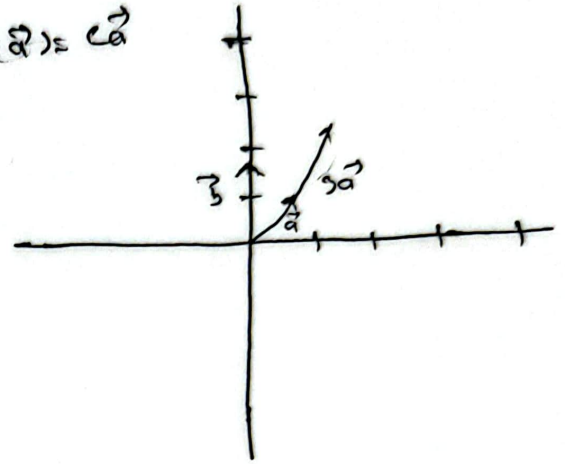
$$c_1 \rightarrow c_n \in \mathbb{R}$$

$$\vec{a} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \vec{b} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

$$3\vec{a} + (2\vec{b}) = \begin{bmatrix} 3-0 \\ 6-0 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$$

$$\text{span}(\vec{a}, \vec{b}) = \mathbb{R}^2$$

$$\text{span}(\vec{a}) = c\vec{a}$$



$$\hat{i} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \hat{j} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\text{span}(v_1, v_2, \dots, v_n) = \{ c_1 v_1 + c_2 v_2 + \dots + c_n v_n \mid c_i \in \mathbb{R} \text{ for } 1 \leq i \leq n \}$$

$$\vec{a} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \vec{b} = \begin{bmatrix} 0 \\ 3 \end{bmatrix} \quad \vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$c_1 \vec{a} + c_2 \vec{b} = \vec{x}$$

$$c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 3 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$1c_1 + 0c_2 = x_1$$

$$2c_1 + 3c_2 = x_2$$

